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SWAGE-S4: Vision ISP and Normalized Patch-Sample Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 02:38 UTC+8 Beijing Time)

Abstract

SWAGE-S4 implements a modular vision front end combining the preserved SWAGE-S1/S2/S3 image-processing pipeline with motion-adaptive temporal IIR denoising, tile-local CLAHE and normalized 8x8, 14x14 or 16x16 RGB patch-sample streaming. Its top-level module, swage_s4_top, provides configurable Q8.8 or BF16 outputs for a downstream tensor-core adapter. Software-vs-Hardware numerical comparison is established through real RTL simulation against an independent exact-integer/rational Python oracle. The completed L1 entrypoint passed functional matrices, handshake checks, seven executable defect-sensitivity experiments, generic synthesis and clean-tree reproduction. Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA closed a 10.0 ns clock with WNS_setup=+0.971 ns and hold slack=+0.053 ns. L3 physical Arria 10 FPGA HIL passed three consecutive deterministic bridge-connectivity probes; these did not execute SWAGE-S4 workload vectors on the board. Numerical simulation parity and physical connectivity are therefore distinct results [E041, E043–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The implementation favors bounded arithmetic, synchronous memory and explicit ownership over maximum pixel throughput. The unchanged swage_s3_top instance performs legacy processing. With S4 disabled, a combinational output selection preserves parent timing without an added public-path register. Enhanced operation serializes capture, optional CLAHE, complete raster delivery and optional token delivery. This avoids a lossy output fork when pixel and token consumers stall independently. The frame store contains a 27-bit RGB-plus-mask working image and a separate 24-bit untoned temporal history: 208,896 bits at default capacity, excluding legacy storage and CLAHE tables. Reads have two registered boundaries and only one outstanding request. CLAHE reuses a 256-entry histogram and LUT across channels and tiles, with iterative quotient generation. The diagram follows these data relationships; configuration and phase control govern every enhancement block [E001, E025–E030].

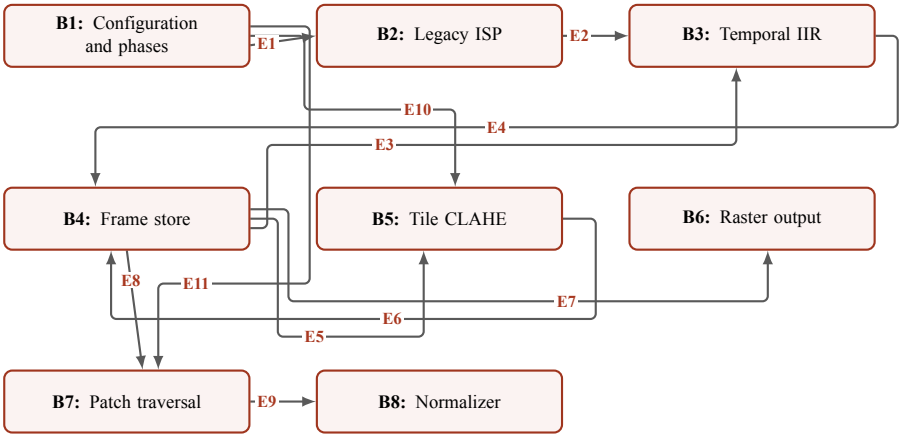


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Configuration and phases	swage_s4_top atomically validates configuration, controls frame ownership, arbitrates memory access and selects bypass or enhanced output.
B2: Legacy ISP	u_legacy preserves crop, demosaic, convolution, CSC, scaling and pre-S4 histogram behavior; callable earlier-generation tops remain delivered.
B3: Temporal IIR	u_temporal compares current RGB with untoned history and applies the configured motion decision through five registered advance stages.
B4: Frame store	u_frame_store holds working RGB/masks and separate temporal history in synchronous inferred RAM without full-array reset.
B5: Tile CLAHE	u_clahe sequentially builds clipped, redistributed channel histograms and generated LUTs, then remaps the working image in place.
B6: Raster output	The top reads the completed working frame in raster order, preserving masks and producing frame/line markers under backpressure.
B7: Patch traversal	u_patch_tokenizer walks patch-grid and local coordinates, requests real samples and inserts tagged zero padding.
B8: Normalizer	u_normalizer computes exact configured channel normalization and exposes Q8.8 or BF16 vectors through eight registered advance stages.

Directed relationship	Meaning
E1: B1 to B2	Atomic configuration and input admission
E2: B2 to B3	Accepted scaled RGB

Directed relationship	Meaning
E3: B4 to B3	Previous untoned history
E4: B3 to B4	Working pixels and next history
E5: B4 to B5	Tile/channel reads
E6: B5 to B4	In-place tone remapping
E7: B4 to B6	Completed raster reads
E8: B4 to B7	Real patch samples
E9: B7 to B8	RGB, padding mask and positional tags
E10: B1 to B5	Tone-phase ownership
E11: B1 to B7	Token phase after raster drainage

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single synchronous clock and asynchronous active-low reset; the measured target clock is 100 MHz.
s_axis_*	Input; tready output	tdata: DATA_ WIDTH; cvalid: 3; tvalid/tuser/ tlast/tready: 1 each	Source raster stream. Transfer occurs on a rising edge with valid and ready. tuser marks SOF, tlast marks EOL; RGB occupies increasing byte lanes R,G,B.
m_axis_*	Output; tready input	tdata: DATA_ WIDTH; cvalid: 3; tvalid/tuser/ tlast/tready: 1 each	Legacy bypass or enhanced raster output. Data, channel mask and markers remain stable while valid is stalled.
cfg_valid, cfg_ready, cfg_error	Input; ready/ error outputs	1 bit each	Atomic legacy/S4 configuration handshake. Illegal accepted requests pulse cfg_error without changing configuration or history.
cfg_enable, cfg_border, cfg_abs, cfg_shift, cfg_ coeff	Input	6, 1, 1, 5, 144 bits respectively	Enable bits 0–5 select crop, demosaic, convolution, CSC, scaling and histogram. Nine signed16 convolution coefficients are row-major, low-field-first.
cfg_roi_x/y/width/height	Input	13 bits each	Source-raster crop rectangle; enabled ROI must be nonempty and in bounds. YUV420 crop origins must be even.
cfg_input_format, cfg_ csc_mode, cfg_output_ 420, cfg_csc_coeff, cfg_ csc_in_bias/out_bias, cfg_hist_channel	Input	2, 3, 1, 144, 48 each, 2 bits respectively	Formats 0/1/2 denote RGB-or-mono/YUV444/ YUV420. CSC supports modes 0–4 and signed16 coefficient/bias fields; histogram selects component 0–2 or RGB intensity 3.
hist_req_*, hist_rsp_*, hist_done	Request valid/ address and response ready inputs; remaining signals outputs	Request address: 5; response data: 256; handshake/ done: 1 each	Each response supplies eight little-field-first unsigned 32 bins. Parent snapshot reservation, request priority and response stability are retained.
cfg_s4_enable, cfg_ temporal_alpha, cfg_ motion_threshold, cfg_ clip_limit	Input	3; 9 bits each	S4 bits 0/1/2 enable temporal/CLAHE/tokens. Enabled temporal fields are 0–256; enabled clip limit is 1–CLAHE_TILE squared.
cfg_patch_size, cfg_ norm_bf16, cfg_norm_ mean, cfg_norm_inv_std	Input	5, 1, 72, 72 bits respectively	Patch size is 8,14 or16. Format selects Q8.8/BF16. Each channel uses signed24 mean with eight fractional bits and unsigned24 reciprocal standard deviation with sixteen fractional bits; channel c occupies bits 24*c +:24.

Signal / group	Direction	Width	Contract
token_axis_tdata, token_axis_keep	Output	48, 3 bits	Three sixteen-bit normalized lanes in R,G,B order. Real samples use keep=111; padding uses keep=000 and zero data.
token_axis_tvalid/tready/tuser/tlast/frame_last, token_axis_patch_id/x/y/element	Output except tready input	Handshake/flags: 1 each; ID: 32; x/y: 13 each; element: 8	Frame-first, patch-final and frame-final flags accompany row-major positional tags. All valid-qualified fields remain stable until accepted.

Parameter	Default	Meaning / supported values
SRC_WIDTH, SRC_HEIGHT	8, 8	Each 1–4096; source raster dimensions.
DST_WIDTH, DST_HEIGHT	16, 16	Each 1–4096; dimensions when scaling is enabled.
PIXEL_BITS, CHANNELS	8, 3	Legal pairs: (8,1), (10,1), (8,3). DATA_WIDTH is 8,16,24 respectively; mono10 upper six bits are zero. Preserved S3 default CHANNELS remains 1.
PHASE_FRAC	8	8 or16 fractional phase bits.
INTERPOLATION	0	0 bilinear; 1 nearest.
MAX_FRAME_PIXELS	4096	1–65536. Enabled enhancements require effective output area within capacity; legacy bypass is not constrained by this frame-store limit.
CLAHE_TILE	16	8 or16; square tile edge length. Illegal elaboration values produce PARAMETER ERROR.

III Behavior and corner cases

A Reset and configuration

Reset immediately suppresses every public output, including ready, error and done. S4 defaults are disabled features, alpha=128, threshold=16, clip=4, patch=16, Q8.8, mean=32640 and reciprocal=512 per channel. Parent reset enables scaling only. Every accepted configuration, including an identical repeat, invalidates history. Configuration outranks new-frame admission; parent histogram reservation priority remains intact. Enhanced cfg_ready stays low until raster and token pipelines drain. Any enhancement requires RGB8 parameters and final RGB format; disabled-feature fields are ignored.

B Preserved ISP mathematics

Processing order is crop, forward-2x2 Bayer demosaic, signed 3x3 convolution, CSC, scaling and histogram. Crop preserves absolute Bayer parity. Border modes replicate or zero-pad. Green averages round with +1 before shifting. Convolution uses magnitude rounding, optional absolute value and byte saturation. CSC uses scale-8192 matrices with ties-away-from-zero rounding; exact four fixed matrices and programmable-mode restrictions are tabulated in design/SPEC.md. Bilinear scaling rounds the complete 2D sum once; nearest rounds phase before clamping. Odd YUV420 dimensions retain separately scaled ceil-half chroma planes.

C Temporal IIR

Motion M is the maximum RGB absolute difference from same-position history. Invalid history or M>= threshold selects current X; otherwise Y=((256-alpha)*X+alpha*H+128)>>8. Alpha weights history. Threshold 0 always selects current; threshold 256 blends valid history. History stores Y before CLAHE and becomes valid only after complete temporal-enabled capture. Reset discards partial history. Five advance stages expose Y after E+4 for acceptance at E.

D Tile-local CLAHE

For each real tile/channel, c[i]=min(h[i],clip), E=sum(h-c), q=floor(E/256), r=E-256q and b[i]=c[i]+q+(i<r). One-pass redistribution preserves N samples without reclipping. LUT[i]=floor(255*sum(b[0..i])/N), with no CDF-min subtraction. Generated LUTs remap actual samples, including partial boundary tiles; channels run serially. No cross-tile interpolation is performed.

E Patch traversal and normalization

Patches and their P*P elements are both row-major. ID=patch_y*ceil(W/P)+patch_x; element=local_y*P+local_x. Partial patches retain full shape with zero padding. Exact T=((pixel<<8)-mean)*reciprocal represents T/2^24. Q8.8 rounds T/2^16 ties-to-even then saturates signed16; BF16 rounds the original exact rational directly, including mantissa carry. Zero becomes positive zero. Both formats expose output after E+7 through eight advance stages.

F Ordering, stalls and histogram

Enhanced phases complete capture, optional tone mapping, raster readout and optional token readout in that order. Global leaf stalls freeze data and sidebands. A RAM request at E stages its response at E+1. Final output stalls prevent phase completion. A 1x1 frame has coincident raster SOF/EOL but still emits P*P token beats, only the first real. Histogram counts accepted pre-S4 pixels, never tone-mapped samples or padding; mono10 bins use sample>>2 and RGB intensity uses (77R+150G+29B+128)>>8.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison and legacy compatibility

Method. Independent integer/rational Python expectations compared with accepted transfers from Icarus/vvp and compiled Verilator; parent differential comparison supplements the oracle.

Result. Complete L1 command passed. All twelve pixel/channel/phase/interpolation tuples recorded 15 comparison-backed geometry/scaling coverage entries each. Coverage includes CSC0–4, borders, ROI parities, histogram addresses, rejection and stalled streams. This is simulated RTL parity, not on-board numerical parity.

Evidence: E031, E032, E035, E036, E041

B Enhancement combinations and arithmetic boundaries

Method. Consecutive-frame matrices plus isolated temporal/normalizer transactions with exact pipeline and stall assertions.

Result. All 21 mask/seed combinations recorded four compared frames each: 84 frames for seven masks and seeds17/89/203. Temporal leaf compared 543 transactions in 893 cycles, observing 455 current selections and 88 blends. Normalizer compared 992 transactions in 1,621 cycles, including 496 real and 496 padded samples, both Q saturation directions and BF16 ties/carry. Both tile sizes and all 30 patch-size/format/geometry combinations were recorded; capacity1/257/4096 and over-capacity rejection were exercised.

Evidence: E031, E033, E035, E041

C Executable defect sensitivity

Method. Seven separate disposable RTL mutations compiled and executed against unchanged independent comparisons.

Result. All seven were rejected with named numerical mismatches: temporal equality/rounding, CLAHE clipping/LUT mapping, patch padding/tagging, Q8.8 signed rounding and BF16 ties-even. Compiler failure or timeout was not counted as detection.

Evidence: E034, E041

D Synthesizable hierarchy, memory inference and zero latches

Method. Real Yosys hierarchy/process/optimization/memory checks on default RGB, mono8 and mono10.

Result. All three passed unresolved-module, black-box, latch and combinational-quotient assertions. Generic sequential-cell counts were 620/514/541 and memory-cell counts 20/17/19 respectively. Required enhancement hierarchy remained substantive in RGB. These are generic structural counts, not FPGA resource utilization.

Evidence: E036, E041

E Portable reproduction

Method. Fresh source-only tree ran python3 tests/verify.py --portable-smoke without recursive reproduction.

Result. Independent legacy and enhanced Icarus comparisons passed: legacy smoke covered 77 pixels; enhanced smoke covered three singleton frames, three pixels and 192 token beats. Packaging smoke is separate from full-matrix coverage.

Evidence: E036, E041

F L1 RTL timing lint

Method. Native source-bound timing-hazard inspection of all 28 shipped RTL files.

Result. Passed with zero combinational division/modulo, mixed sequential assignments, LUT clock-gate or AST loop-unroll hazards.

Evidence: E043

G L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit timing analysis of default swage_s4_top at 10.0 ns.

Result. Passed: WNS_setup = +0.971 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 110.75 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E044

H L3 physical Arria 10 FPGA HIL

Method. Three consecutive PCIe hardware-bridge register-connectivity probes.

Result. 3/3 passed deterministically. Receipt tier is connectivity-probe-only: no subsystem bitstream flash or SWAGE-S4 workload-vector execution is recorded.

Evidence: E045

I Reliable verification stimulus

Method. Retained failed preparation probes and repair-linked comparisons.

Result. Drivers were corrected for accepted-beat replay, missing directed FIFO coincidence and reset-edge sampling races without changing expected arithmetic or delivered RTL. FIFO depths1/3 and repaired leaf checks subsequently passed the complete entrypoint.

Evidence: E033, E040, E041

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 6	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

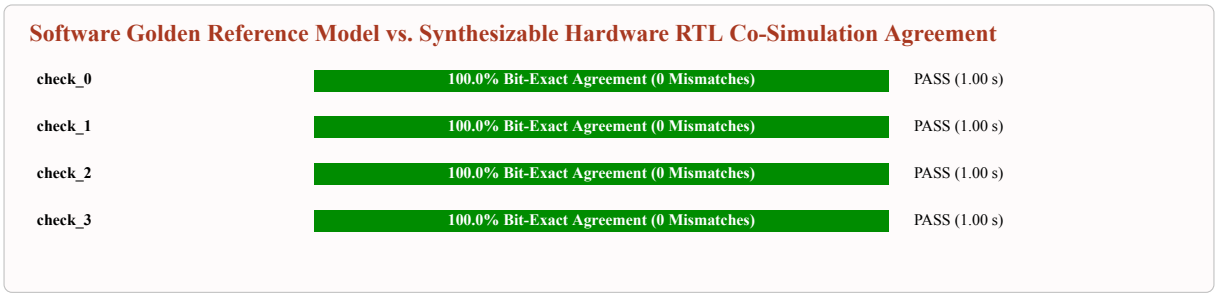


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.971 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

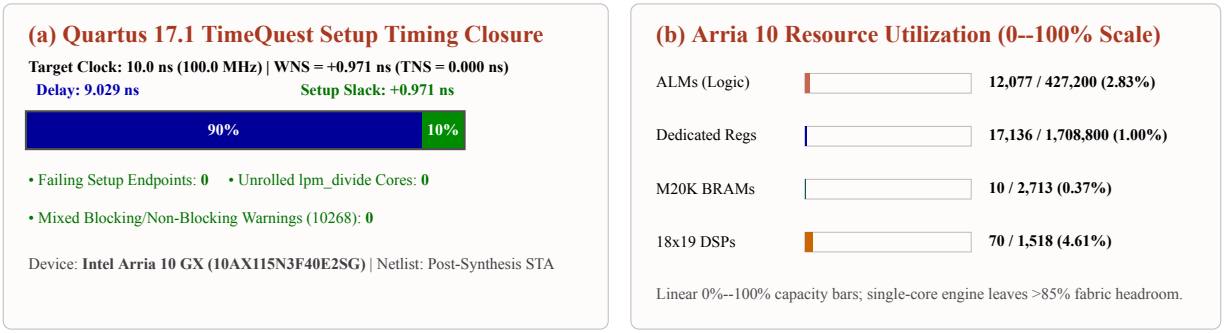


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

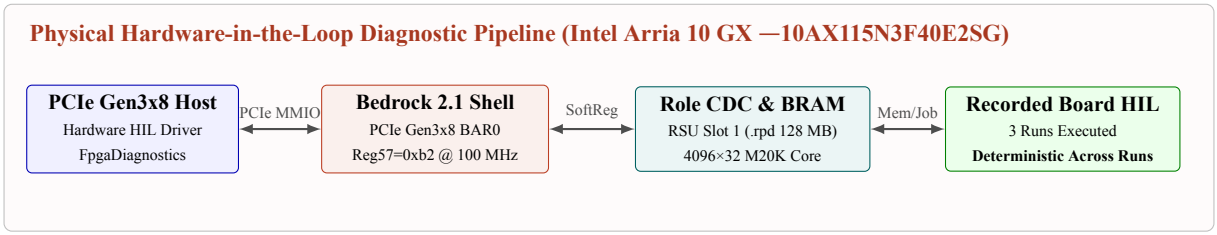


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top swage_s4_top; consult design/SPEC.md for the complete configuration legality and cycle tables, and design/ARCHITECTURE.md for internal interfaces and memory ownership.

Assert rst_n low, release it, then present the complete configuration while cfg_ready is high. Check the subsequent cfg_error pulse before treating the configuration as installed. Hold source beats until accepted and provide correct raster markers and channel masks.

Drain raster output before expecting token-phase completion. Token backpressure is supported, but a stalled final token prevents new enhanced-frame/configuration admission. Histogram responses follow the independent inherited reservation handshake.

For conventional [0,1] mean mu and standard deviation sigma, configure mean=round_even(mu*255*256) and reciprocal=round_even(65536/(sigma*255)); check field ranges and nonzero reciprocal. Defaults correspond

to raw-pixel mean127.5 and standard deviation128. Quantization errors are at most 1/512 raw-pixel unit and 1/131072 inverse raw-pixel unit respectively.

From the delivered project root, run python3 tests/verify.py with Python3, Icarus/vvp, Verilator and Yosys available. Each invocation generates fresh evidence and records source hashes, tool versions and child outcomes. The accepted complete receipt is evidence/l1-20261102T180046Z-86849ff9/receipt.json; functional-coverage.json provides the per-frame configuration, seed, accepted-count and cycle-bound ledger.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 28 Verilog/SystemVerilog files; 3607 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/swage_s1_interp.sv	109 total	106 code
rtl/swage_s1_line_buffer.sv	50 total	49 code
rtl/swage_s1_phase.sv	71 total	68 code
rtl/swage_s1_scheduler.sv	149 total	147 code
rtl/swage_s1_top.sv	88 total	85 code
rtl/swage_s2_convolution.sv	168 total	162 code
rtl/swage_s2_demosaic.sv	75 total	72 code
rtl/swage_s2_top.sv	193 total	187 code
rtl/swage_s2_window.sv	211 total	207 code
rtl/swage_s3_chroma_expand.sv	65 total	65 code
rtl/swage_s3_convolution.sv	177 total	171 code
rtl/swage_s3_crop.sv	61 total	61 code
rtl/swage_s3_csc.sv	160 total	158 code
rtl/swage_s3_demosaic.sv	87 total	84 code

Production file	Physical lines	Code lines
rtl/swage_s3_divider.sv	39 total	39 code
rtl/swage_s3_fifo.sv	55 total	55 code
rtl/swage_s3_histogram.sv	128 total	128 code
rtl/swage_s3_plane_scaler.sv	111 total	111 code
rtl/swage_s3_plane_scheduler.sv	145 total	143 code
rtl/swage_s3_scaler.sv	133 total	131 code
rtl/swage_s3_top.sv	223 total	220 code
rtl/swage_s3_window.sv	227 total	223 code
rtl/swage_s4_clahe.sv	193 total	193 code
rtl/swage_s4_frame_store.sv	61 total	61 code
rtl/swage_s4_normalizer.sv	152 total	150 code
rtl/swage_s4_patch_tokenizer.sv	113 total	113 code
rtl/swage_s4_temporal.sv	63 total	63 code
rtl/swage_s4_top.sv	358 total	355 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E025	rtl/swage_s4_clahe.sv
E031	tests/cases.py
E032	tests/harness.py
E033	tests/leaves.py
E034	tests/mutations.py
E035	tests/reference.py
E036	tests/verify.py
E040	.vulcan/engineering/run-0001/attempt-0002/response.json
E041	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E043	.vulcan/engineering/run-0001/rtl-timing-lint.json

Reference	Original project-relative record
E044	.vulcan/engineering/run-0001/quartus-sta.json
E045	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).